Canvas and SVG

Canvas and SVG (Scalable Vector Graphics) are both popular technologies used in web development for creating graphics and visualizations, but they have different approaches and use cases.

Introduction to Canvas

It is a pixel-based drawing surface that is widely used for creating dynamic and interactive content on the web.

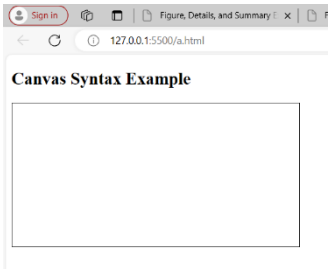
Syntax:

```
<canvas id="myCanvas" width="width_value" height="height_value"
style="border:1px solid #000;"></canvas>
```

Attributes:

- **id:** A unique identifier for the canvas.
- **width:** The width of the canvas.
- **height:** The height of the canvas.
- **style:** Additional styles for the canvas, such as borders or background color.

HTML Code	Output

`<body>`
`<h2>Canvas Syntax Example</h2>`
`<canvas id="myCanvas" width="400" height="200" style="border:1px solid #000;"></canvas>`
`</body>`
	

In this example, a canvas element with an id of "myCanvas" is created with a width of 400 pixels and a height of 200 pixels. The style attribute adds a black border around the canvas for visibility.

Scalable Vector Graphics (SVG)

Scalable Vector Graphics (SVG) is a widely adopted XML-based markup language for describing two-dimensional vector graphics. SVG graphics are resolution-independent and can be scaled to any size without losing clarity.

Syntax:

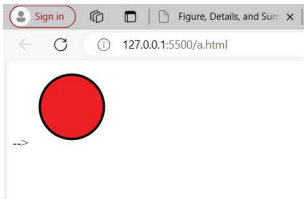
```
<svg width="width_value" height="height_value" xmlns=
"http://www.w3.org/2000/svg">
   <!-- SVG content goes here -->
</svg>
```

Attributes:

- **id**: Provides a unique identifier for the SVG element.
- **class**: Assigns one or more class names to the SVG element.
- **style**: Specifies inline styles for the SVG element.
- **fill**: Specifies the fill color for the graphic element.
- **stroke**: Defines the color of the element's outline.
- **stroke-width**: Sets the width of the element's outline.
- **viewBox**: Defines the position and dimensions of the SVG content in the user coordinate system.
- **preserveAspectRatio**: Specifies how the SVG content should be scaled and positioned within the viewport.

Note: Attributes specific to graphic elements like <circle>, <rect>, <line>, etc. are also utilized within SVG markup.

HTML Code	Output

`<body>`
`<svg width="200" height="100" xmlns= "http://www.w3.org/2000/svg">`
`<circle cx="50" cy="50" r="40" fill="red" stroke="black" stroke-width="3" />`
`</svg>`
`</body>`
	

The cx and cy attributes specify the center coordinates, r is the radius, and fill, stroke, and stroke-width control the appearance.